TRAVENTURE

Travel and Tourism Management System

Lab Report: Use Case Diagram, Activity Diagram & User Stories

Submitted by:
Rehnuma Salsabil-23524202021
Sumaiya Hossain-23524202089
Shreya Sarker-23524202109
Nuzhat Saima-23524202145

1. Objective(s)

- To understand the concept of use case diagrams in UML.
- To identify actors and use cases of the TRAVENTURE system.
- To represent system functionality graphically using UML diagrams.
- To draw the activity diagram showing the complete user interaction flow.
- To define user stories for each actor in the system.

2. Problem Analysis

TRAVENTURE supports three distinct user roles — Admin, Registered User, and Guest User — each with different access levels and capabilities across modules: flight booking, hotel booking, travel packages, chatbot, and admin dashboard.

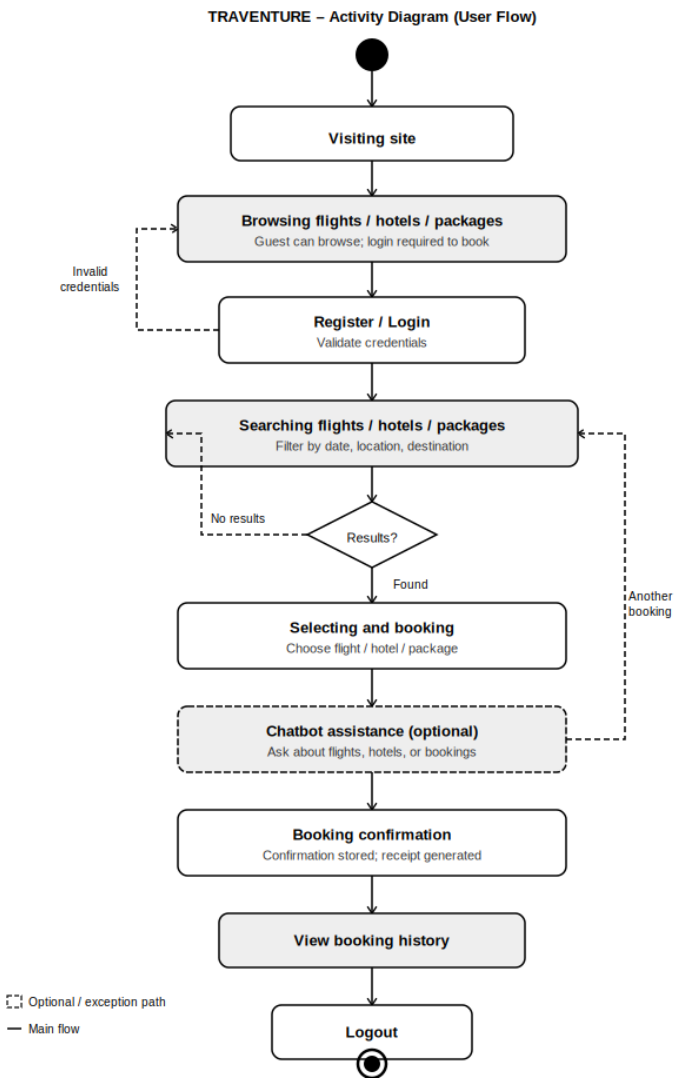

Figure 2: Activity Diagram of TRAVENTURE Travel and Tourism Management System

3. Use Case Diagram

A Use Case Diagram is a type of UML (Unified Modeling Language) diagram that represents the functional requirements of a system. It shows the interaction between actors and system use cases.

3.1 Key Components

Component	Symbol	Description
Actor	Stick Figure	A user or external entity interacting with the system
Use Case	Oval / Ellipse	A specific function or service the system provides
System Boundary	Rectangle	Defines the scope of the system; all use cases inside belong to the system
Association	Solid Line	Shows interaction between an actor and a use case

Component	Symbol	Description
Include	Dashed Arrow «include»	Mandatory inclusion of one use case within another
Extend	Dashed Arrow «extend»	Optional extension of a base use case
Generalization	Solid Arrow	Inheritance relationship between actors or use cases

4. Methodology

Step 1: Identify the System Scope

- System: TRAVENTURE Travel and Tourism Management System
- Modules: User Management, Flight, Hotel, Travel Package, Booking Management, Chatbot

Step 2: Identify All Actors

- Admin – Manages all system data; full access to backend
- Registered User – Authenticated user; can search, book, and view history
- Guest User – Unauthenticated visitor; limited browsing only

Step 3: Identify All Use Cases

- Extracted from SRS Section 3.1 (Functional Requirements)
- Grouped by actor and module

Step 4: Establish Relationships

- Association: All actors connected to their respective use cases
- Include: Login «include» Validate Credentials; Book Flight «include» Confirm Booking
- Extend: View Receipt «extend» Book Flight

5. Actors and Use Cases of TRAVENTURE

5.1 Actor Description

Actor	Role	Permissions
Admin	System manager with full backend access	Manage users, flights, hotels, packages, bookings
Registered User	Authenticated platform user	Register, login, search, book, view history, use chatbot
Guest User	Unauthenticated visitor	Browse limited info; must register to book

5.2 Complete Use Case List

#	Use Case	Description	Actor(s)
1	Register	New user creates an account with name, email, password	Guest → Registered User
2	Login	Authenticate using valid credentials	Registered User, Admin
3	Logout	Securely end the current session	Registered User, Admin
4	Browse Flights	View available flights (limited for guest)	Registered User, Guest
5	Search Flights	Filter flights by source, destination, date	Registered User
6	Book Flight	Select and confirm a flight booking	Registered User
7	Search Hotels	Find hotels by location and date	Registered User
8	Book Hotel	Reserve an available hotel room	Registered User
9	Browse Travel Packages	View available travel packages	Registered User, Guest
10	Book Travel Package	Purchase and confirm a travel package	Registered User
11	View Booking History	Access personal past and current bookings	Registered User
12	Use Chatbot	Get assistance via chatbot interface	Registered User
13	Manage Users	View, update, and delete user accounts	Admin
14	Manage Flights	Add, update, delete flight records	Admin
15	Manage Hotels	Add, update, delete hotel records	Admin
16	Manage Travel Packages	Add, update, delete travel packages	Admin
17	Manage Bookings	View and manage all booking records	Admin
18	View Dashboard	Access admin backend with system overview	Admin

6. Use Case Diagram

The Use Case Diagram below shows the interactions between the three actors: Admin, Registered User, and Guest User, and the system's functional modules within the TRAVENTURE system boundary.

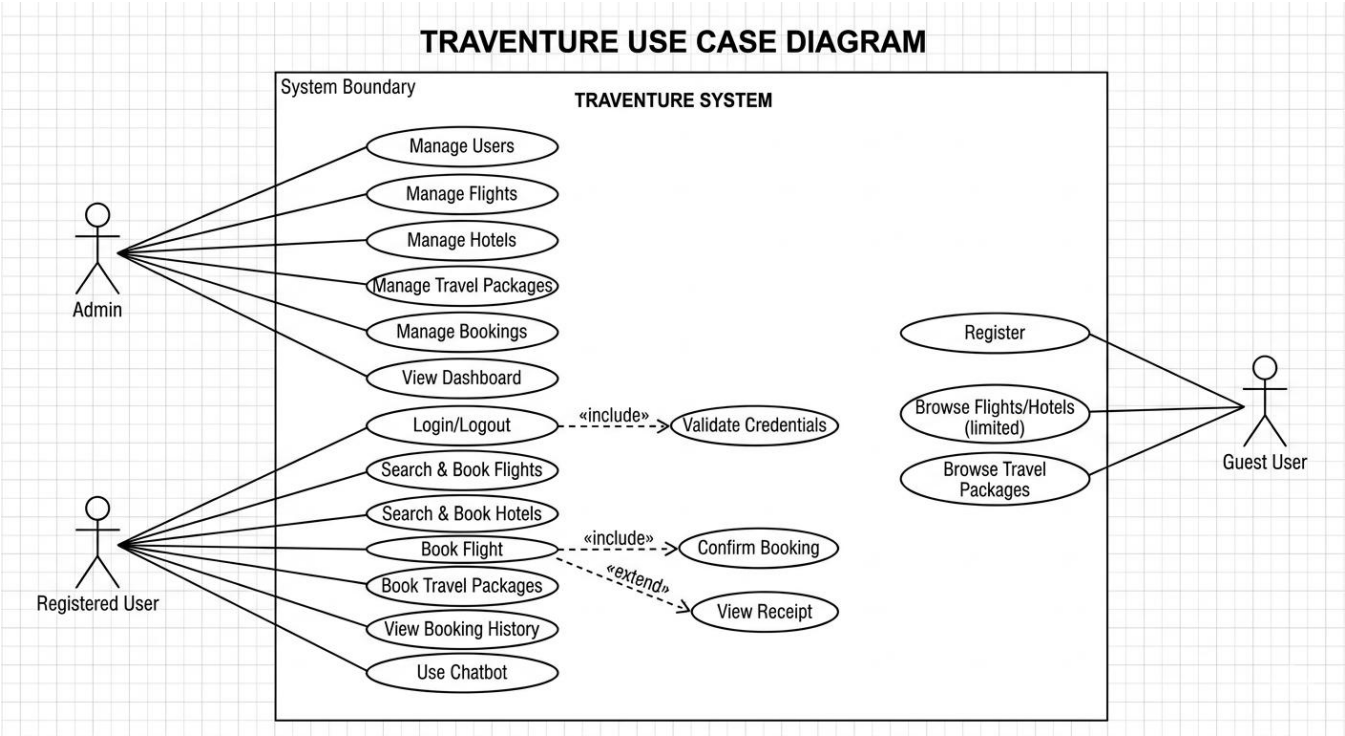

Figure 2: Use Case Diagram of TRAVENTURE Travel and Tourism Management System

7. Activity Diagram – TRAVENTURE User Flow

An Activity Diagram is a UML behavioral diagram that models the workflow and sequence of actions in a system. It shows decision points, parallel flows, and loops.

The diagram below illustrates the complete user flow within TRAVENTURE, from visiting the site to logout. Solid arrows represent the main flow; dashed arrows represent optional or exception paths (invalid credentials, no results found, another booking).

7.1 Activity Diagram Key Steps

#	Activity	Description
1	Visiting site	User navigates to TRAVENTURE; guests can browse without login
2	Browsing flights/hotels/packages	Guest or user views available travel options; full booking requires login
3	Register / Login	User creates an account or authenticates; loops back if credentials are invalid
4	Searching	User filters flights, hotels, or packages by date, location, and destination
5	Results decision	If no results found, user is returned to the search step to refine criteria
6	Selecting and booking	User chooses a flight, hotel, or travel package and confirms the booking
7	Chatbot assistance	Optional: user interacts with chatbot; can loop back to make another booking
8	Booking confirmation	System stores confirmation and generates receipt
9	View booking history	User reviews all past and current bookings
10	Logout	Session is securely terminated

8. User Stories for TRAVENTURE

8.1 Registered User Stories

#	User Story
1	As a user, I want to register an account so that I can access TRAVENTURE's booking features.
2	As a user, I want to log in with my credentials so that the system can verify my identity.
3	As a user, I want to search for flights by source, destination, and date so that I can find suitable travel options.
4	As a user, I want to view flight details (airline, time, price) so that I can compare and choose the best option.
5	As a user, I want to book a selected flight so that I can confirm my travel plans.
6	As a user, I want to search for hotels by location and date so that I can find accommodation at my destination.
7	As a user, I want to book an available hotel so that I have confirmed accommodation for my trip.
8	As a user, I want to browse travel packages so that I can find all-in-one travel deals.
9	As a user, I want to book a travel package so that I can organize my complete trip in one step.
10	As a user, I want to view my booking history so that I can track all my past and upcoming trips.
11	As a user, I want to use the chatbot so that I can get quick answers about flights, hotels, and bookings.

#	User Story
12	As a user, I want to log out securely so that my account remains protected after my session.

8.2 Admin User Stories

#	User Story
1	As an admin, I want to securely log in to the dashboard so that I can manage the system safely.
2	As an admin, I want to manage user accounts so that I can maintain the integrity of the user database.
3	As an admin, I want to add new flights so that users always have up-to-date flight options.
4	As an admin, I want to update or delete flight records so that outdated data is removed.
5	As an admin, I want to add and manage hotels so that users have accurate accommodation choices.
6	As an admin, I want to manage travel packages so that users can access current deals.
7	As an admin, I want to view and manage all bookings so that I can resolve issues and maintain records.
8	As an admin, I want to access the backend dashboard so that I can monitor system performance.

8.3 Guest User Stories

#	User Story
1	As a guest, I want to browse available flights and hotels so that I can explore options before registering.
2	As a guest, I want to register an account so that I can unlock full booking capabilities.
3	As a guest, I want to view travel packages so that I can assess whether TRAVENTURE meets my needs.